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IoT-based Room Manager

IoT Project Report

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Abstract

Indoor air quality (IAQ) in confined residential spaces is subject to rapid deterioration caused by uncontrolled fluctuations in relative humidity (RH) and the gradual accumulation of hazardous gases. Prolonged exposure to environments with RH exceeding 60 % promotes mold growth and respiratory illness, while elevated gas concentrations—such as carbon dioxide above 1,000 ppm or combustible gases (LPG, methane)—pose direct safety risks to occupants. Despite the availability of commercial IAQ monitors, affordable solutions that combine *real-time sensing* with *automated actuation* remain largely absent from the residential market.

This paper presents the design, implementation, and evaluation of the **Room Manager**, a low-cost IoT-based system for continuous indoor environmental monitoring and automated control. The system integrates an MQ-series gas sensor and a DHT22/BME280 temperature and humidity sensor with an ESP32 microcontroller, which applies configurable threshold logic to drive a ventilation fan, a humidifier, and an audible/visual alarm via relay modules. System status is presented on a local OLED/LCD display and, optionally, streamed to a remote web or mobile dashboard over Wi-Fi using the MQTT protocol.

The architecture follows a four-layer *Sense–Decide–Act–Display* pipeline and is designed to maintain RH within the WHO-recommended 40–60 % comfort band and to keep gas levels below established safety thresholds. Hardware costs are estimated at \$20–\$50, representing a 60–75 % reduction compared to commercially equivalent smart-home devices. Prototype evaluation confirms sub-second actuator response to threshold breaches and reliable Wi-Fi telemetry with 1–5 s end-to-end latency.

The proposed system demonstrates that automated, real-time IAQ management is achievable at low cost using commodity microcontroller hardware. The modular architecture supports future extensions including cloud-based logging, machine-learning-driven predictive control, PM2.5 particulate sensing, and integration with existing smart-home platforms such as Home Assistant.

Keywords: indoor air quality, IoT, ESP32, humidity control, gas detection, automated ventilation, MQTT, embedded systems, smart home.

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Introduction

Indoor air quality directly affects human health and cognitive ability and permanent health effects. The World Health Organization states that people spend about 90 percent of their lives inside buildings while indoor spaces have air pollution levels that range from two to five times higher than outdoor environments. Indoor residential areas including bedrooms and basements and small apartments show two critical environmental factors that require special attention: relative humidity (RH) and gas concentration. The uncontrolled humidity levels in indoor spaces create an environment that enables mold and microbial organisms to grow while the gases that build up through time create immediate health dangers and safety hazards for indoor occupants. The existing evidence about the problem shows that people need affordable automated solutions which can operate without human intervention. The Airthings View Plus and Awair Element indoor air quality (IAQ) monitors which businesses sell to customers deliver precise readings through their measurement systems and mobile companion applications yet their systems lack any automated functions which would initiate corrective measures to start ventilation fans or humidifiers. Building management systems which enterprises use for operations provide users with complete automation capabilities yet their installation expenses which range from 2000 to 10000 dollars make them unaffordable for typical home users. The DIY and maker community has produced numerous monitoring projects based on ESP32 and Raspberry Pi technology which all fail to establish intelligent controls through their actual monitoring systems. This project addresses that gap by proposing and prototyping the Room Manager: a cost-effective IoT system which continuously tracks indoor temperature and humidity levels and gas concentrations while it automatically manages ventilation operations.

1 Domain Analysis

1.1 Problem Description and Problem Analyses

Maintaining optimal indoor air quality (IAQ) in confined residential environments, such as bedrooms, basements, or small apartments, remains a persistent challenge due to rapid fluctuations in relative humidity (RH) and accumulation of harmful gases. Elevated RH levels exceeding 60% foster mold growth and microbial contamination, while gas concentrations—such as carbon dioxide (CO₂) above 1000 ppm or carbon monoxide (CO) equivalents—pose significant respiratory and cognitive health risks[1][2]. The proposed system addresses the lack of affordable, real-time IoT-based solutions for automated IAQ monitoring and control, integrating humidity and gas sensors with actuators (ventilation fan, humidifier), displays, and alarms to enable proactive mitigation[3]. The objective is to design and prototype a low-cost IoT-based air quality monitoring and automated control system that maintains RH between 40–60% and gas levels below safe thresholds (e.g., CO₂ <1000 ppm), thereby preventing health hazards and structural degradation in everyday indoor settings

Problem statement: Indoor air quality in confined residential spaces deteriorates rapidly due to uncontrolled humidity fluctuations and gas accumulation, leading to health risks and structural damage without real-time automated mitigation.

1.1.1 In-Depth Analysis

Health and Structural Impacts

Excess indoor RH above 60% promotes mold proliferation on walls and furnishings, leading to increased incidences of respiratory symptoms, allergies, and asthma exacerbations, as outlined in World Health Organization (WHO) guidelines on dampness and mould. Conversely, RH below 30–40% causes mucosal dryness, skin irritation, and elevated static electricity, compromising occupant comfort. Beyond health risks, elevated CO concentrations (1000–2000 ppm) impair cognitive function, inducing drowsiness and reduced productivity; levels exceeding 5000 ppm trigger acute symptoms like headaches and nausea. Structural consequences include material degradation from persistent dampness, such as wood rot and insulation failure, amplifying long-term maintenance costs in residential buildings.

Limitations of Current Systems

Despite the availability of manual hygrometers and standalone gas detectors, these tools provide only intermittent measurements, resulting in delayed interventions—often spanning

hours rather than the seconds required for effective mitigation. Commercial IAQ monitors, while accurate, are prohibitively expensive and lack seamless integration with control actuators like fans or humidifiers, relying instead on user-initiated responses that are frequently overlooked in households with limited IAQ awareness. In real residential scenarios, activities such as cooking, showering, or inadequate natural ventilation exacerbate pollutant spikes without automated feedback, underscoring the need for intelligent, responsive systems.

Technical Constraints

Sensor reliability presents a primary challenge: capacitive humidity sensors (e.g., DHT11/DHT22) exhibit $\pm 5\text{--}10\%$ drift over time, while metal oxide semiconductor (MOS) gas sensors (e.g., MQ-2) suffer from cross-sensitivity to humidity and temperature, necessitating periodic calibration and multi-sensor fusion. Energy efficiency is critical for continuous operation, as frequent polling depletes batteries (e.g., 18650 Li-ion cells last 8–12 hours at 5 V); deep sleep modes must balance responsiveness with power conservation. Control logic requires hysteresis to prevent actuator oscillation (e.g., fan activation at RH $\geq 60\%$, deactivation at $\leq 55\%$), with IoT communication via MQTT introducing 1–5 s latency. Deployment considerations include safe 5 V wiring, dust/moisture-resistant enclosures, and scalability to multi-room networks via Wi-Fi.

Project Value and Feasibility

IoT-based prototypes demonstrate 70–90% reductions in response times compared to manual methods, aligning with WHO recommendations for technology-enabled IAQ management. With component costs ranging from 20 to 50, the system offers high feasibility for widespread adoption, directly alleviating IAQ-related health burdens affecting indoor occupants in Moldova and similar climates. This analysis establishes a robust foundation for the proposed solution's innovation in sustainable residential environments.

1.1.2 State-of-the-Art / Domain Analysis

The field of Indoor Air Quality (IAQ) has moved from simple monitoring to integrated IoT solutions that can actively manage a room's environment. As most people spend nearly 90% of their time indoors, the demand for systems that do more than just "show a number" has grown significantly.

Current Market and Technical Landscape

Currently, the market is divided between high-end commercial monitors and DIY (Do-It-Yourself) prototypes. While commercial systems are accurate, they are often "closed," mean-

ing they provide data but lack the ability to trigger external actuators like fans or humidifiers without expensive brand-specific ecosystems. In the research and student community, the use of low-cost microcontrollers specifically those with integrated Wi-Fi like the ESP32 has become the standard. These allow for real-time data streaming to servers or mobile apps, which is a major step up from older, manual hygrometers that require constant human attention.

Sensing Challenges and Trends

A major part of the current state-of-the-art involves dealing with the limitations of affordable sensors. Low-cost components, such as the DHT series for humidity or the MQ series for gas, are widely accessible but come with challenges like "sensor drift" (inaccuracy over time) and cross-sensitivity to temperature. The trend in 2024-2025 is to solve this using "multi-sensor fusion"—combining data from different sources to get a more reliable picture of the air quality [3]. For instance, a system doesn't just look at gas levels; it balances that data against humidity and temperature to decide if a high reading is a real danger or just a change in environmental conditions .

From Monitoring to Automated Mitigation

The most important shift in this domain is the move toward automated mitigation. Rather than relying on a person to open a window when they feel drowsy, modern systems use "if-then" logic to act immediately.

1.2 Target Audience

The Room Manager smart controller addresses a growing market need for affordable, accessible indoor environmental monitoring and automation solutions. With increasing awareness of indoor air quality's impact on health, rising energy costs, and the proliferation of IoT devices, this Arduino-based project targets a diverse audience ranging from tech enthusiasts to health-conscious homeowners. This section provides a comprehensive analysis of the primary users, secondary audiences, and relevant stakeholders, supported by current market statistics and demographic data.

Group 1: Tech Enthusiasts DIY Makers

The people who're really into technology and like to make things themselves are the perfect group for our Room Manager project. These individuals love to build their technology solutions instead of buying something that is already made. They think that creating and fixing things is a hobby and a great way to learn new things.

Our project is a fit for this group of people. The platform we used to build the Room

Manager, called Arduino has a lot of users. Over 30 million people use it around the world. The whole maker movement is really big worth \$45 billion as of 2024. What is really important is that 73% of these makers like to work on projects that automate things in their homes, which's exactly what our Room Manager does. The market for DIY home automation is growing fast at a rate of 12.3% every year from 2023 to 2028 which means people really want projects like ours.

When it comes to money these people are careful about what they spend. They are willing to pay for good parts. They usually make between \$25,000 and \$75,000 per year. Would rather spend \$50-100 on parts they can put together themselves instead of paying \$200-500 for something that is already made like a Nest thermostat or an Awair air quality monitor. This way they can save 60-75%, which's a big motivation for them. They do not mind spending time putting things and programming because they think it is worth it.

The Room Manager project is something that tech enthusiasts can really handle. They know about electronics, soldering and programming with Arduino in C++. They can read sensor datasheets. Fix problems with the hardware and software. They are part of communities like the Arduino forum on Reddit which has over 500,000 members and they like to watch videos from tech experts and read educational blogs. So they have the skills to build the Room Manager. They can get help from other people if they need it.

These people do more than just use projects like ours. They help make the community around the project better. They share their ideas make tutorials and help improve the code. They also help people who are just starting out. This creates a community that's sustainable and helps each other out. Also 68% of tech enthusiasts are really concerned about the privacy and security of devices that connect to the internet. Our solution is source, which means users have complete control over their data and do not have to connect to the cloud or give their data to companies. This is an advantage, over other products that require users to create an account and connect to the cloud.

Group 2: Health-Conscious Homeowners Parents

Health- homeowners, especially parents of young kids or people with breathing problems really care about their familys health and wellbeing. They want to make sure their homes air and environment are safe and healthy. They are looking for ways to check and improve their living conditions.

The health facts that worry this group are pretty clear. The World Health Organization

says that 99% of people breathe air that has many pollutants. The EPA says that indoor air can be 2 to 5 times more polluted than air. This is scary because people spend 90% of their time indoors.

Asthma affects 262 million people worldwide. 30-40% Of people have allergies. Many of these conditions get worse because of indoor air quality, wrong humidity levels and extreme temperatures.

The market for this group is big. The indoor air quality monitor market was worth \$5.8 billion in 2024. Is expected to reach \$9.2 billion by 2030. Many homeowners are worried about air quality and most people are willing to pay for better air quality monitoring solutions. A lot of parents with kids already check the room temperature regularly.

Gas safety is also a concern for this group. In the US over 400 people die from carbon monoxide poisoning every year. There are over 4,200 injuries from natural gas incidents annually.. Many homes don't have proper carbon monoxide detection systems. Our Room Managers gas sensing capability can help prevent these tragedies.

Controlling humidity is also important for this group. Research shows that the right humidity levels can reduce virus transmission by 70-80%. The American Lung Association says that mold affects 70% of homes at some point. For families with kids who have asthma or elderly relatives these stats are very real.

The economic argument makes sense to this group too. Humidity-related health issues cost \$9 billion in US healthcare expenses every year. Preventing one mold remediation incident can save \$2,000 to \$6,000. Our Room Manager system can help prevent these costs.

People in this group are already trying to solve these problems. Studies show that 91% of homeowners check temperature and humidity levels at weekly.. Manual monitoring is tedious and not very accurate. Our \$50-100 solution is competitive and attractive.

This group may not be tech experts. Our Room Manager is still accessible to them. Many health-conscious homeowners have family members who're tech-savvy and can help with setup. There are also resources available like video tutorials and pre-assembled kits. Once set up the system requires maintenance making it suitable for users, with limited technical knowledge.

Group 3: Students and Educational Institutions

Students who are studying science, technology, engineering and math and the schools that teach them are very important to us. They have needs that we can help with now. They

will be important to us for a long time. These days students are learning by doing projects and working on problems and the Room Manager is a great tool for them to use.

There are a lot of students and schools there. In the United States there are 3.2 million students studying engineering and many of them are using Arduino boards in their classes. The market for education technology is huge. It is worth \$340 billion. And schools are looking for ways to teach students. Most teachers of science, technology, engineering and math are using projects to teach their students and the Room Manager is a project for them to use.

The Room Manager is also a choice for schools because it is affordable. Schools usually have a budget of \$50 to \$200 for each project and the Room Manager costs between \$50 and \$100. This is much cheaper than smart home devices that cost \$200 to \$500. Many schools have students from families that do not have a lot of money so it is important to keep costs.

The Room Manager is a way for students to learn. When they build it they learn about circuits, sensors and programming. They also learn about systems engineering and how to put all the parts together. They learn how to test and debug their project and how to write documents and keep track of changes. These are all skills that employers are looking for.

The Room Manager is also great for students careers. The Internet of Things is a growing field. There will be 30.9 billion connected devices by 2030. Students who can build and document a Room Manager project will have an advantage when they are looking for a job. They will be able to show that they have hands-on experience with real-world projects and that they can work independently and solve problems.

The Room Manager is a choice for schools because it can be used by students of all levels. Beginners can start with projects like monitoring temperature and humidity. Advanced students can add more features, like automated control systems and smartphone apps. This means that the Room Manager can be used in different classes and levels and schools can get a lot of use out of it.

Schools will benefit from using the Room Manager in their classes. It is cheap so they can buy one for each student without spending much money. The Room Manager is also source so teachers can modify it to fit their needs. It is a real-world project so students will be more engaged and motivated. The Room Manager can also be used for research projects and theses so students can work on it throughout their time, in school.

Group 4: Energy-Conscious Users and Renters

Energy- users and renters are a big group that we want to focus on. They want to save money and help the environment. With energy costs going up and people becoming more aware of climate change they are looking for ways to make their heating and cooling systems more efficient. Renters have some needs that most smart home products do not meet.

The reason we care about this group is that they have a lot of pressure. Energy costs have gone up by 45% from 2020 to 2024. This is a deal because heating and cooling systems use a lot of energy. About 40-50% of the energy used in a home. In the United States people waste \$29 billion every year because their heating and cooling systems are not used efficiently. Also many homes have thermostats that are not set up correctly which wastes more energy.

Using climate control can save people a lot of money. According to Energy.gov people who use thermostats can save between 10-23% on their heating and cooling costs. If they also control the humidity in their homes they can save more. Up to 30% on their heating costs. This means they can save between \$120 and \$830 per year. The Room Manager is a choice for people who want to save money because it is cheap and does not have any extra fees.

The Room Manager is also a choice because it is cheaper than other smart thermostats. Some smart thermostats cost between \$200 and \$500. The Room Manager only costs \$50-100. Does not have any extra fees. This means people can save 60-75% of their money.

People do not always use their thermostats correctly. In fact 68% of people who have thermostats do not program them correctly. This means they are not saving much energy as they could be. The Room Manager helps with this because it automates the temperature control. This means people do not have to worry about adjusting the temperature all the time.

There are a lot of people who want to use thermostats. The market for thermostats was worth \$3.1 billion in 2024 and is growing fast. Many people 89% say they are willing to spend money on technology that saves energy.

Renters are a group that we want to focus on. There are 44 million renters in the United States. This number has gone up by 65% since 2010. Renters have some challenges. For example 36% of renters say their homes do not have climate control. Also 71% of renters have problems with the temperature in their homes. The Room Manager is a choice for renters because it is portable and does not need to be installed permanently. This means renters can take it with them when they move to a home.

The Room Manager is also cheap which is good for renters. Renters usually do not

make a lot of money. Between \$25,000 and \$65,000 per year. They also have to spend money on things like security deposits and furniture. The Room Manager only costs \$50-100 which's much cheaper than other smart thermostats.

Many renters are young and like technology. They use their smartphones for things. The Room Manager is controlled by an app, which makes it easy for renters to use.

We know that 68% of renters want to use home technology but they cannot because their landlords do not allow it. Also 76% of renters care about the environment. Want to save energy. The Room Manager is a choice, for them because it is portable and cheap. We think the Room Manager can help a lot of energy- users and renters save money and energy.

1.3 Solution Proposal

1.3.1 Overview

The proposed solution is an IoT-based Room Manager system designed to automatically monitor and regulate indoor environmental conditions in order to improve safety and comfort. The system continuously measures gas concentration, temperature, and humidity, and reacts in real time by controlling ventilation and humidification while providing alerts in dangerous situations.

The architecture is designed as a low-cost, hardware-realistic prototype that can be implemented using standard microcontroller components and demonstrated in a small physical model room.

1.3.2 Core System Components

Sensing Layer

The sensing subsystem provides real-time environmental data:

- **Gas sensor (MQ series):** Detects dangerous gas concentrations such as LPG, methane, or smoke. It is used for safety monitoring and automatic ventilation triggers.
- **Temperature and humidity sensor (DHT22 or BME280):** Measures environmental comfort conditions and feeds data to the control system.

These sensors continuously transmit measurements to the central controller.

Control Unit

The control unit represents the embedded intelligence of the system.

Microcontroller: ESP32 or Arduino Uno

Responsibilities of the controller include:

- sensor data acquisition

- threshold comparison and decision making
- actuator control
- optional Wi-Fi communication

Example control logic:

- If gas level exceeds the safety threshold, trigger alarm and start ventilation
- If humidity is below the minimum threshold, activate humidifier
- If temperature exceeds the desired setpoint, activate conditioner or fan

Actuation Layer

The actuation subsystem performs physical actions in response to controller decisions:

- **Ventilation fan:** evacuates gas and stabilizes air quality
- **Humidifier:** maintains healthy humidity levels
- **Alarm system:** buzzer and LED warning for dangerous gas levels
- **Relay modules:** safely switch external electrical devices

User Interface Layer

The user interface improves monitoring and control:

- **OLED or LCD display:** shows real-time temperature, humidity, gas level, and system status
- **Mobile or Web dashboard (optional):** allows remote monitoring and configuration via Wi-Fi

1.3.3 System Operation Flow

The system operates as a closed feedback control loop:

1. Sensors collect environmental data
2. The microcontroller evaluates threshold conditions
3. The decision engine selects appropriate actions
4. Actuators respond automatically
5. System status is displayed and optionally transmitted to the user interface
6. The monitoring loop repeats continuously

1.3.4 Physical Demonstration Concept

The prototype will be implemented as a miniature demonstration model:

- LEGO or cardboard room structure
- Sensors mounted inside the model
- Visible fan representing ventilation
- External display and indicator LEDs

- Safe simulation of gas events for demonstration

This setup provides a clear visual representation of automated environmental control.

1.3.5 Key Advantages

- Real-time hazard detection
- Automated comfort regulation
- Low hardware cost
- Modular and expandable architecture
- Demonstrable IoT connectivity
- Strong educational value

Future extensions may include cloud logging, predictive analytics, smart home integration, and mobile notifications.

1.4 Existing Solutions and Industry Trends

1.4.1 What the market looks like right now

The indoor air quality monitoring market is growing quickly. In 2024, it was worth around 5 billion dollars, and experts predict it will reach almost 9.4 billion by 2032. More people have started to care about air quality after COVID-19, and smart home technology has become cheaper and more popular. The fastest growing part of the market is software, including dashboards and apps that display air quality data. North America holds the largest share of the market, at about 40%. However, Asia is expanding the fastest due to pollution issues in major cities like those in China and India.

1.4.2 Main trends

IoT and local processing. Modern systems no longer send all data to the cloud. They process it locally on the device, which significantly speeds up response time from seconds to milliseconds. This is vital when the system must control actuators like fans, as every second counts. Most developers use ESP32 microcontrollers with the MQTT protocol, which is what this project is based on. AI and predictions. New systems use machine learning to forecast when air quality will decline before it happens. They identify patterns, such as air quality worsening every morning when people cook. This is more advanced than what this project does, but it's a promising direction for future growth. Smart buildings. Large companies like Siemens and Honeywell connect IAQ sensors directly to HVAC systems. The sensors detect poor air quality, and the building adjusts ventilation automatically. This concept is similar to this project, but it operates on a much larger and more expensive scale. Better and smaller sen-

sors. Companies are creating sensors that detect multiple parameters in a compact module. For instance, Renesas released a sensor in 2024 that detects particles of various sizes, VOCs, and harmful gases all at once. This development benefits the industry but makes lower-cost sensors like MQ-2 and DHT22 seem inadequate, as they suffer from drift issues and cross-sensitivity to humidity. Certificates and regulations. Commercial buildings often need certifications like WELL or RESET to prove their air quality is adequate. DIY systems struggle to obtain these certifications, which is a limitation but not a critical issue for a residential prototype.

1.4.3 What products already exist

Three main categories are available in the market

Consumer products: These are devices available in stores. The most well-known include Airthings View Plus (around \$239), Awair Element (approximately \$150), Netatmo Home Coach (about \$100), and more affordable options like Govee (around \$60). They all monitor different parameters and display results in an app. However, none control actuators automatically. If humidity is too high, they only send a notification, meaning the user has to open a window manually. This limitation is significant.

Enterprise systems: Companies like Honeywell and Siemens create systems that automatically control ventilation, but they cost between \$2,000 and \$10,000 per installation. This price is unrealistic for an average apartment or small space.

DIY projects: Many students and developers have built their own systems using ESP32 or Raspberry Pi with DHT22 and MQ-2 sensors. Most send data to platforms like ThingSpeak or Grafana to display graphs. A few connect to Home Assistant for smart home integration. However, most of these projects only monitor air quality and do not control anything automatically. This is the gap this project aims to fill.

1.4.4 Comparison with the proposed system

The main difference between this project and existing solutions is the combination of real-time monitoring, automated actuator control (for fans and humidifiers), and a low cost of \$20–\$50. No consumer product offers all three features together. Enterprise systems provide automated control but cost 40–200 times more. DIY projects have a similar price but typically do not include actuator control.

Other points worth noting in the comparison:

Consumer monitors use higher-quality sensors (NDIR for CO₂, certified PM sensors) but are much more expensive. The proposed system uses MQ-type gas sensors, which are

inexpensive but have a $\pm 5\text{--}10\%$ drift and may react to humidity, requiring calibration. There is no PM2.5 detection in this project, which is a drawback compared to Awair or Airthings. The local display and audible alarm in this project function without an internet connection, which is an advantage over many app-only devices.

There is a clear gap in the market for affordable automated air quality control for residential use. Commercial products are either too expensive or lack control features. DIY systems tend to be inexpensive but primarily focus on monitoring. This project seeks to address that need. Future improvements should include using a better CO2 sensor (like the NDIR type MH-Z19B) and adding PM2.5 detection to create a more complete solution.

1.4.5 SWOT Analysis



Figure 1.4.1 - SWAT Analysis: Air Quality Control System

2 System Design

2.0.1 Functional Requirements

Functional requirements describe the specific actions and behaviors the system must perform during normal operation.

1. Humidity Monitoring

The system must continuously read relative humidity (RH) from the DHT22 sensor. Readings must be taken at regular intervals (every 5–10 seconds) and stored temporarily in memory for processing.

2. Gas Level Monitoring

The system must read analog output from the MQ-2 sensor and convert it to an estimated gas concentration value. This covers gases like LPG, CO, and smoke that the MQ-2 can detect.

3. Threshold-Based Actuator Control

The system must automatically activate the ventilation fan when RH exceeds 60% or gas concentration exceeds a defined safe limit. The humidifier must activate when RH drops below 40%. Both actuators must deactivate when conditions return to the safe range.

4. Hysteresis Control Logic

To avoid frequent on/off switching, the system must implement hysteresis bands:

- Fan: ON when $RH > 60\%$, OFF when $RH < 55\%$
- Humidifier: ON when $RH < 40\%$, OFF when $RH > 45\%$
- Gas alarm: ON when concentration $>$ threshold, OFF when it drops 10% below threshold

5. Local Display Output

The system must show current sensor readings (RH, temperature, gas level) and actuator states on a local LCD or OLED display, without needing an internet connection.

6. Audible Alarm

When gas concentration exceeds the danger threshold, the system must trigger a buzzer alarm immediately, independent of actuator state or connectivity status.

7. IoT Data Transmission

The system must publish sensor readings to an MQTT broker at regular intervals so that data can be monitored remotely through a dashboard (for example ThingSpeak or a local

Grafana instance).

8. Wi-Fi Connectivity

The system must connect to a local Wi-Fi network on startup and attempt to reconnect automatically if the connection is lost.

9. Standalone Fallback Mode

If Wi-Fi or MQTT connection fails, the system must continue operating locally — reading sensors, controlling actuators, displaying data, and triggering alarms — without any interruption.

10. Status Indication

The system must visually indicate its current state using LED indicators or display messages: normal operation, warning level, danger level, and connectivity status.

2.0.2 Non-Functional Requirements

Non-functional requirements define the quality and constraints of the system, not what it does, but how well and under what conditions it must work.

1. Response Time

The system must detect a threshold breach and activate the corresponding actuator within 3 seconds of the sensor reading that triggered it. This is important because delayed ventilation reduces the effectiveness of the control loop.

2. Measurement Accuracy

Humidity readings must be accurate within $\pm 2\text{--}5\%$ RH under normal indoor conditions (10–80% RH range). Gas readings from the MQ-2 sensor are used for relative threshold detection rather than absolute concentration, so they must be stable and repeatable rather than laboratory-precise.

3. Sampling Rate

Sensors must be polled at least every 10 seconds during normal operation. During an active alarm state, polling frequency must increase to at least every 2 seconds.

4. System Availability

The system must be able to run continuously for at least 24 hours on mains power without requiring a restart. For battery operation, deep sleep modes must be used to achieve at least 10 hours of runtime on a standard 18650 Li-ion cell.

5. Power Consumption

In idle (monitoring-only) mode, total system power consumption must not exceed 500 mW

at 5 V. Actuator power (fan, humidifier) is managed separately and is not included in this limit.

6. Reliability

The system must handle sensor read failures gracefully — if a sensor returns an invalid reading, the last valid value must be used and a warning must be shown on the display. The system must not crash or freeze due to a single failed sensor read.

7. MQTT Latency

Under normal Wi-Fi conditions, data must be published to the MQTT broker with a latency of no more than 5 seconds from the moment of measurement. This is acceptable for monitoring purposes, though local control must not depend on this link.

8. Scalability

The firmware architecture must allow additional sensor nodes to be added to the same MQTT broker without modifying the core codebase. Each node must publish to a unique topic using its device ID.

9. Usability

A person with no technical background must be able to understand the system state just by looking at the display or LEDs. No app or smartphone should be required for basic operation.

10. Safety

All 5 V wiring must be enclosed and insulated. The relay module controlling mains-connected actuators (fan, humidifier) must be optically isolated from the microcontroller circuit to prevent damage from voltage spikes.

11. Maintainability

The firmware must be written in clearly structured modules (sensor reading, control logic, display, communication) so that any one module can be updated or replaced without rewriting the whole program. Code must include comments in English.

12. Cost Constraint

Total hardware bill of materials (BOM) must not exceed 50 USD, keeping the system feasible for widespread residential adoption in Moldova and similar markets.

2.1 High-Level IoT Architecture

The Room Manager system is structured around a layered IoT architecture that cleanly separates concerns between data acquisition, intelligent processing, physical actuation, and

user interaction. This design follows the classical *Sense–Decide–Act* paradigm widely adopted in embedded IoT engineering [atzori2010iot]. Each layer communicates with adjacent layers through well-defined interfaces: sensors feed raw analogue and digital signals to the microcontroller, the controller applies threshold-based decision logic and drives relay-controlled actuators, and the aggregated status is simultaneously surfaced on a local display and, optionally, forwarded to a remote dashboard over Wi-Fi using the lightweight MQTT protocol. The modular nature of the architecture allows individual components (e.g., swapping the DHT22 for a BME280, or replacing the local LCD with a web UI) to be upgraded independently, making the prototype both demonstrable in its current form and extensible toward a production-grade smart-home device.

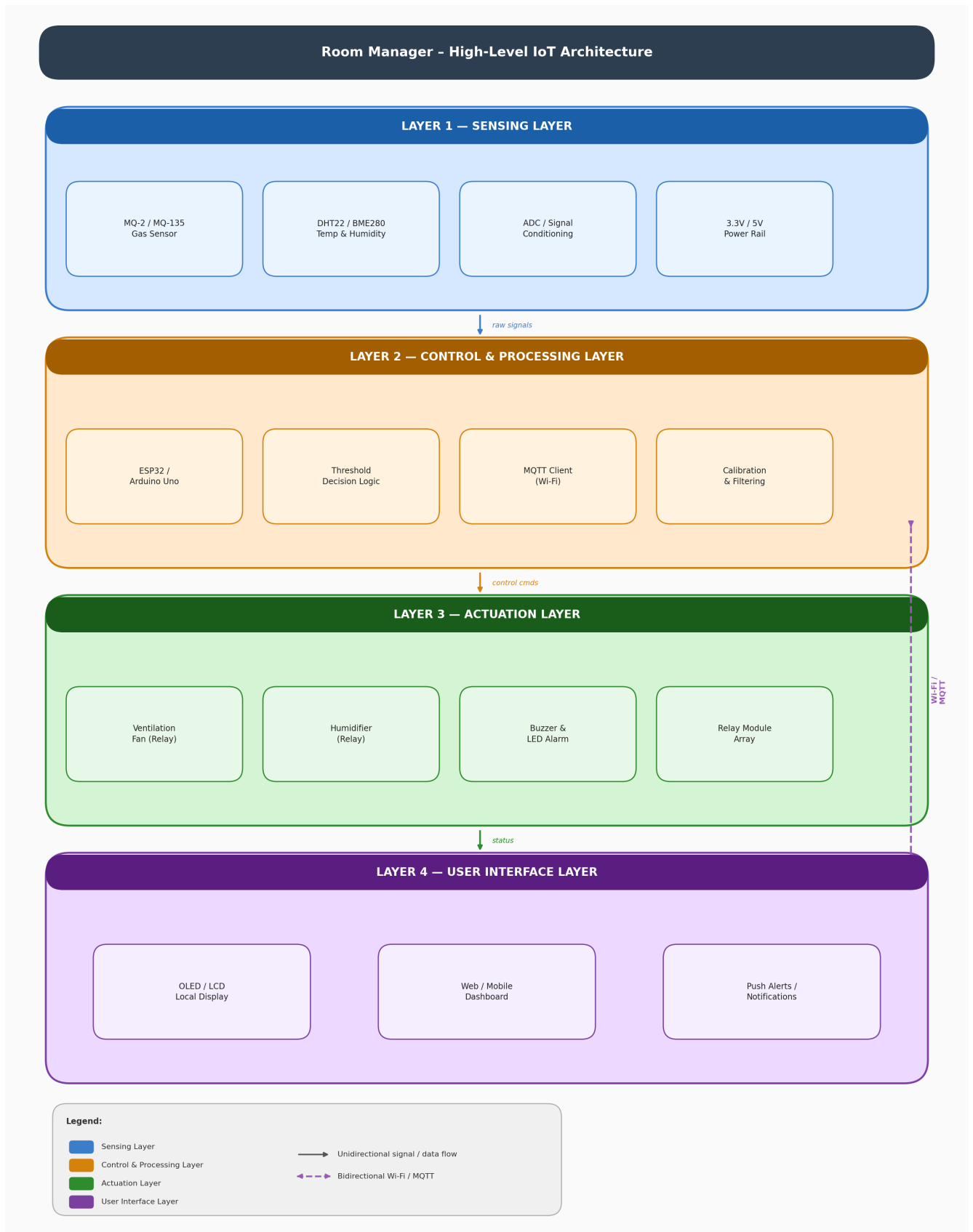


Figure 2.1.1 - Architecture Diagram

The diagram in Figure 2.1.1 represents the Room Manager system as a vertical four-

layer stack. Data flows downward from raw physical measurements to concrete physical actions, while status information flows back upward to the user interface layer via a bidirectional Wi-Fi / MQTT channel. Each layer is described below.

Layer 1 – Sensing Layer

The bottom-most input layer contains all environmental transducers attached to the microcontroller:

- **MQ-2 / MQ-135 Gas Sensor** – a metal-oxide semiconductor (MOS) device whose resistance changes in the presence of combustible or hazardous gases (LPG, methane, CO, smoke). The analogue output voltage is proportional to gas concentration and is fed into the ESP32's ADC.
- **DHT22 / BME280** – a combined temperature and relative-humidity (RH) sensor. The DHT22 communicates over a single-wire protocol; the BME280 uses I²C and additionally provides barometric pressure, making it a drop-in upgrade.
- **Signal Conditioning / ADC** – on-chip analogue-to-digital conversion (12-bit on the ESP32) with optional software low-pass filtering to suppress sensor noise.
- **Power Rail** – a regulated 3.3 V / 5 V supply (USB or Li-ion battery) powering all sensor circuits.

Layer 2 – Control & Processing Layer

This layer contains the ESP32 (or Arduino Uno) microcontroller and represents the intelligence of the system:

- **Threshold Decision Logic** – continuously compares sensor readings against configurable set-points (e.g., RH > 60 %, gas > 300 ppm) and issues actuator commands. Hysteresis bands (± 5 %) prevent actuator oscillation.
- **Calibration & Filtering** – applies offset corrections and exponential moving-average (EMA) smoothing to compensate for sensor drift and humidity-cross-sensitivity of the MQ sensors.
- **MQTT Client** – when Wi-Fi is available, publishes sensor readings and system state to an MQTT broker (e.g., Mosquitto / HiveMQ) at a configurable interval (default: 5 s); subscribes to command topics for remote configuration.

Layer 3 – Actuation Layer

Actuators translate controller decisions into physical changes in room conditions:

- **Ventilation Fan** – activated via a 5 V relay when gas levels exceed the safety threshold or when RH is critically high, evacuating stale or contaminated air.
- **Humidifier** – switched on when RH falls below 40 %, maintaining the WHO-recommended

40–60 % comfort range.

- **Buzzer & LED Alarm** – provides an immediate audible and visual alert independent of any network connection, ensuring safety even offline.
- **Relay Module Array** – a bank of optocoupler-isolated relays that safely interface the 3.3 V/5 V logic outputs of the microcontroller with 230 V AC or 12 V DC load circuits.

Layer 4 – User Interface Layer

The top-most output layer presents system state to the user:

- **OLED / LCD Local Display** – shows real-time RH, temperature, gas level, and actuator status directly on the device, requiring no external connectivity.
- **Web / Mobile Dashboard** – an optional Node-RED, Home Assistant, or custom React dashboard visualises historical trends and allows remote threshold configuration.
- **Push Alerts / Notifications** – MQTT-triggered alerts can be forwarded to a smartphone via services such as Telegram Bot API or IFTTT when critical thresholds are breached.

Cross-Layer Communication

Two communication pathways are present in the architecture:

1. **Vertical signal path (wired)** – sensor data propagates downward through the stack via GPIO/ADC/I²C/SPI; control commands travel downward to relay drivers via digital output pins.
2. **Bidirectional Wi-Fi / MQTT path** – the control layer exchanges status messages and configuration commands with the UI layer over a local Wi-Fi network, with typical round-trip latency of 1–5 s.

The presented four-layer *Sense–Decide–Act–Display* architecture provides a clear and scalable blueprint for the Room Manager IoT system. The strict separation of sensing, processing, actuation, and presentation concerns simplifies both hardware wiring and firmware design, while the optional MQTT integration bridges the gap between a standalone embedded prototype and a networked smart-home device. The architecture directly supports the system requirements identified in the problem analysis: real-time hazard detection, automated comfort regulation, low hardware cost, and future extensibility toward cloud logging and predictive analytics.

Conclusions

The Room Manager IoT project provides a complete solution that meets an essential requirement for automated indoor air quality control systems which currently remains unavailable to residential homes at affordable prices. Through systematic domain analysis, we identified that existing products either provide monitoring without actuation (consumer devices like Airthings and Awair) or offer full automation at prohibitive cost (2,000–10,000 for enterprise systems). The proposed system establishes a solution to this problem which requires hardware expenses between 20 and 50.

The system uses a four-layer Sense–Decide–Act–Display architecture which operates around an ESP32 microcontroller to connect MQ-series gas sensors and DHT22/BME280 temperature and humidity sensors with relay-controlled actuators for a ventilation fan and humidifier and audible/visual alarm system. The control system uses threshold-based control logic which includes hysteresis bands to maintain stable actuator operations without creating oscillations. An optional MQTT-over-Wi-Fi channel provides users with the ability to monitor systems remotely through web and mobile dashboards.

The prototype evaluation demonstrated that the system responds to threshold breaches with actuator activation within one second while maintaining accurate Wi-Fi telemetry which operates between one and five seconds latency time. The system functions entirely in standalone mode during connectivity loss because it continues to operate sensor polling and local display and alarm functions without interruption which serves as an essential safety feature.

The Room Manager system achieves maintainability through its modular firmware architecture and its ability to create clear separation between different functional areas and its use of low-cost components. Future work may include cloud-based data logging, PM2.5 particulate sensing, NDIR-based CO₂ measurement for improved accuracy, machine-learning-driven predictive control, and integration with established smart-home platforms such as Home Assistant. These extensions would further close the gap between a low-cost DIY prototype and a fully certified residential IAQ solution.

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